

DANIEL T. KETTERER

SOFTWARE DEVELOPER & SYSTEMS ANALYST

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PROFESSIONAL SUMMARY

Mathematician (M.S.) and full-stack developer with defense R&D experience and an inactive Top Secret clearance. Background spans systems-level engineering (Linux, C/C++, static analysis), enterprise integration security (OAuth 2.0, Salesforce, Azure), and applied mathematics (optimization, machine learning, control). Comfortable translating complex theoretical and domain requirements into clean, maintainable software.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, MATLAB, Java, SAP ABAP, Apex

Systems & Security: Linux (shell, systems-level), static analysis / call-graph tracing, OAuth 2.0 / PKCE, refresh-token rotation, secret management (Azure Key Vault), CI/CD (GitHub Actions)

Frameworks & Tools: Flask, Flet, Git/GitHub, PostgreSQL, MySQL, Postman/Newman, JMeter, Jest, Google Maps API

Machine Learning & Applied Math: CNNs, clustering, reinforcement learning, ant colony optimization, control theory

EXPERIENCE

IT Analyst, McGohan Brabender

Jan 2025 - Feb 2026

Dayton, OH

- Secured Salesforce-to-SharePoint/GitHub integrations with OAuth 2.0 (PKCE and refresh-token rotation), Connected Apps, and secret management via Azure Key Vault.
- Built file-driven integration workflows with webhook reliability and end-to-end audit/alerting through Azure Log Analytics.
- Developed and maintained Salesforce automations (Flows; Apex triggers, queueables, batch jobs) and optimized SOQL.

Math Intelligence Engineer, Mercor

Feb 2025 - present

Remote

- Apply advanced mathematics to evaluate and strengthen the mathematical reasoning of large language models.

Principal Developer, SafePlate

Sep 2023 - Sep 2024

- Built a full-stack, cross-platform allergy-conscious restaurant finder app with Flet (Python UI), a Flask backend, and a MySQL database.
- Integrated the Google Maps API for location services and delivered persistent storage and API integration end to end.

Quality Assurance Engineer, PayTile

Jun 2022 - Apr 2023

- Performance- and load-tested the PayTile API for 5,000 concurrent users with Redline13 and JMeter.
- Built an automated test suite (Postman, Newman, GitHub Actions) for CI/CD and wrote TypeScript/Jest unit tests.

Graduate Researcher, Joint Inference and Control

Sep 2019 - May 2022

Wright State University / Air Force Research Laboratory (AFRL) - Dr. Luther Palmer

- Developed autonomous control strategies for military UAV surveillance in anti-access / area-denial (A2/AD) environments in MATLAB.
- Implemented an Ant Colony Optimization algorithm and applied CNNs, clustering, and reinforcement learning to the control research.

Federal System Developer Analyst, Accenture Federal Services

Mar 2016 - Jul 2017

U.S. Department of Defense - GFEBS

- Maintained GFEBS, a large legacy Army financial system, writing SAP ABAP to diagnose and repair defects.
- Built the Call Trace Tool (CTT), a static-analysis utility that recursively traces external calls and builds a program call graph; adopted on a second federal contract (DEAMS).

Software Tester, Accenture Federal Services

Jun 2015 - Jan 2016

U.S. Department of Education

- Verified data consistency across databases with Linux shell and PostgreSQL; traced data flow to identify high- and low-performing schools for a DoE reporting site.

INDEPENDENT RESEARCH

- Quantitative Measurement of Human Flourishing**
- 2026 - present
- Developing "f," a general-factor measurement construct modeling flourishing as systemic coupling strength across life domains, grounded in psychometric factor theory.
 - Authored a pre-registered Global Flourishing Study analysis plan and built simulation studies to test the construct's structure and identifiability.

SECURITY CLEARANCE

Top Secret (held 2016, inactive), U.S. Department of Defense.

EDUCATION

- M.S., Mathematics - Wright State University, Dayton, OH**
- 2020
- B.S., Mathematics, Magna Cum Laude - Schreiner University, Kerrville, TX**
- 2015
- GPA 3.7